Adaptation Helps, but Discharge Summaries Are Not Ready for Low-Supervision Use.

Nina Alvarez, Robert Chen.
Safe Clinical NLP Initiative.

Abstract.

Domain adaptation improved discharge summary generation, but residual hallucination risk remained too high for low-supervision deployment. Factual precision improved from 0.76 to 0.82, yet clinically significant hallucinations still appeared in 6.1 percent of drafts. Human review remained necessary because one in five summaries required material correction.

1. Introduction.

Average gains can hide dangerous residual errors. Rare failures matter in discharge communication.

This report focuses on deployment relevance. Only outcome statements are emphasized.

2. Methods.

Held-out drafts were reviewed by clinicians and pharmacists. Severe errors received explicit adjudication.

The analysis emphasized unsupported medications, follow-up timing, and missing precautions. Section prose was not scored separately.

3. Results.

Domain adaptation improved factual precision from 0.76 to 0.82 and reduced unsupported follow-up instructions by 29 percent across the evaluation set.

Hallucination risk remained too high for low-supervision deployment, because clinically significant hallucinations still appeared in 6.1 percent of drafts after adaptation.

Human review was still required, since 21 percent of summaries needed material factual correction before release to the care team or patient portal.

Table 1. Safety-oriented evaluation after adaptation.

System	Factual precision	Severe hallucinations	Needs correction
Generic	0.76	9.0%	31%
Adapted	0.82	6.1%	21%
Adapted + checklist	0.83	5.8%	18%

4. Discussion.

Adaptation clearly helped, but the remaining high-severity error rate was still incompatible with low-supervision use in discharge workflows.

The practical implication is that adaptation supports reviewer efficiency rather than reviewer removal, especially in medication and follow-up sections.

5. Conclusion.

Adaptation improved discharge summary quality, but the resulting system was not ready for low-supervision deployment and still required human review. This is a qualified positive result rather than a deployment endorsement. A limitation is that the adjudication protocol weighted severe factual errors more heavily than stylistic completeness.

References.

[1] S. Hall and E. Ross. Rare failure analysis for clinical generation. Safety in NLP, 2024.

[2] K. Young. Hallucination audits in discharge summaries. Journal of Clinical Language Safety, 2023.

[3] M. Iyengar and T. Cole. Human review requirements for medical text generation. Care Delivery Informatics, 2024.

[4] R. Moss. Severity-weighted metrics for factual generation. Measurement for Trustworthy AI, 2022.

[5] L. Brennan et al. Medication section errors in generated notes. Clinical Documentation Studies, 2023.

[6] P. Vora. Unsafe optimism in average-score reporting. Reliable Generative Systems, 2024.

[7] J. Simon and A. Park. Deployment thresholds for high-risk language applications. Medical Governance Review, 2022.